

LYDIA FIKIRE

Full-Stack Software Engineer | Digital Agriculture | Digital Health

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with years of experience delivering impactful digital solutions across public-sector agriculture and health systems in Ethiopia. Proven expertise in designing, developing, and deploying end-to-end web applications, ERP systems, and data platforms that serve thousands of users. Passionate about leveraging technology to modernize government services and improve lives through Digital Agriculture and Digital Health innovations.

WORK EXPERIENCE

Full-Stack Software Engineer

June 2025 – June 2026

Development Gateway (An IREX Venture) – Seconded to the Ministry of Agriculture, Ethiopia

- Designed and developed DOVAR, an animal disease registration and tracking system with automated reporting, deployed for use by thousands of farmers and district agricultural specialists (DAs) across Ethiopia.
- Built a Planning & Performance Management System enabling each state ministry office to set annual targets, track performance indicators, and automatically generate performance reports submitted to the Ethiopian Parliament.
- Delivered comprehensive training and ongoing user support to all Ministry staff and affiliated organizations on deployed systems.
- Developed an ICT Support Ticketing System allowing ministry users to submit IT requests and receive structured assistance from ICT staff.
- Created a Digital Budget Management System to monitor budget allocations and expenditures across ministry departments.
- Performed SharePoint customization, configuration, and end-user training to improve internal collaboration and document management.
- Worked on DHIS2 integration to collect agricultural data from kebele level (Ethiopia's lowest administrative unit), enabling data-driven policy decisions.
- Tested, validated, and documented 50+ REST API endpoints using Postman and Swagger/OpenAPI across all deployed systems (DOVAR, PPMS, Budget Management System, ICT Ticketing System), reducing production errors and speeding up integration across the systems' shared dashboard.
- Set up OpenTelemetry for application monitoring, helping maintain 80% system uptime and enabling rapid detection and resolution of errors.

Software Engineer – Digital Health

October 2024 – June 2025

John Snow, Inc. (JSI) – Seconded to Ministry of Health, Ethiopia

- Provided technical support for DHIS2 data element creation using JSON, including data analysis and interactive dashboard configuration for national health monitoring.
- Contributed to improving interoperability between DHIS2 and the eCHIS (electronic Community Health Information System) platform.
- Co-developed the Digital Health Project Inventory (DHPI) system, enabling the Ministry of Health to register, quality-assure, and certify digital health solutions nationwide.
- Assisted in designing and developing the National Health Data Dictionary (NHDD) Application to standardize national health data definitions across all digital health projects.

KEY PROJECTS

Planning & Performance Management System — Ministry of Agriculture, Ethiopia

Stack: Django · React · Postgres

- Built a nationwide system enabling state ministry offices to set annual targets and track performance against key indicators.
- Automated performance report generation for direct submission to the Ethiopian Parliament, replacing a manual process.

- ▶ Delivered training and user support to all ministry staff and affiliated organizations across Ethiopia.
- ▶ Tested and documented all API endpoints using Postman; monitored uptime and error logs with OpenTelemetry to ensure reliability for nationwide use.

Digital Budget Management System — *Ministry of Agriculture, Ethiopia*

Stack: Django · React · Postgres

- ▶ Built a digital platform to track budget allocations and expenditures across ministry departments, improving financial transparency.
- ▶ Enabled real-time monitoring of spending against approved budgets, reducing manual reporting overhead for finance teams.

Digital Health Project Inventory Management System (DHPI) *Ministry of Health, Ethiopia*

Stack: Laravel  Vue.js

- ▶ Designed and developed a centralized platform for the Ministry of Health to register, manage, and track digital health projects across Ethiopia.
- ▶ Implemented a structured quality assurance workflow enabling systematic testing and certification of health technology solutions before national deployment.
- ▶ Provided end-to-end project lifecycle tracking, giving ministry stakeholders real-time visibility into the status and compliance of all registered digital health initiatives.

EDUCATION

Bachelor of Science in Information Systems

2021 – 2024

Addis Ababa University, Addis Ababa, Ethiopia

CGPA: 3.75 / 4.00 — **Graduated with Distinction**

TECHNICAL SKILLS

Web Development: Laravel, JavaScript, Vue.js, React, Django, C#

Mobile Development: Flutter, React Native

Database Management: MySQL, PostgreSQL

Platforms & Tools: DHIS2, Microsoft SharePoint, Odoo ERP, eCHIS

Other: REST APIs, JSON, Postman, Swagger/OpenAPI, OpenTelemetry, Data Analysis, Dashboard Design, System Integration, User Training

LANGUAGES

Amharic: Native | **English:** Fluent

CERTIFICATIONS

Data Science Foundations – Level 1 — *IBM*

Data Science 101 — *IBM*

CCNA: Enterprise Networking, Security, and Automation — *Cisco*

REFERENCES

Anteneh Legesse

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Ministry of Agriculture, Ethiopia

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